

```
#include <stdio.h>
int main()
{
    int a[100],n,k,i,j,temp;
    printf("enter the size of array");
    scanf("%d",&n);
    printf("enter the elements of array");
    for (i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    printf("enter the value of k");
    scanf("%d",&k);
    for (i=1;i<=k;i++){
        temp=a[n-1];
        a[n-1]=a[n-2];
        a[n-2]=a[n-3];
        a[n-3]=a[n-4];
        a[n-4]=a[n-5];
        a[n-5]=a[n-6];
        a[n-6]=a[n-7];
        a[n-7]=a[n-8];
        a[n-8]=a[n-9];
        a[n-9]=a[n-10];
        a[n-10]=a[n-11];
        a[n-11]=a[n-12];
        a[n-12]=a[n-13];
        a[n-13]=a[n-14];
        a[n-14]=a[n-15];
        a[n-15]=a[n-16];
        a[n-16]=a[n-17];
        a[n-17]=a[n-18];
        a[n-18]=a[n-19];
        a[n-19]=a[n-20];
        a[n-20]=a[n-21];
        a[n-21]=a[n-22];
        a[n-22]=a[n-23];
        a[n-23]=a[n-24];
        a[n-24]=a[n-25];
        a[n-25]=a[n-26];
        a[n-26]=a[n-27];
        a[n-27]=a[n-28];
        a[n-28]=a[n-29];
        a[n-29]=a[n-30];
        a[n-30]=a[n-31];
        a[n-31]=a[n-32];
        a[n-32]=a[n-33];
        a[n-33]=a[n-34];
        a[n-34]=a[n-35];
        a[n-35]=a[n-36];
        a[n-36]=a[n-37];
        a[n-37]=a[n-38];
        a[n-38]=a[n-39];
        a[n-39]=a[n-40];
        a[n-40]=a[n-41];
        a[n-41]=a[n-42];
        a[n-42]=a[n-43];
        a[n-43]=a[n-44];
        a[n-44]=a[n-45];
        a[n-45]=a[n-46];
        a[n-46]=a[n-47];
        a[n-47]=a[n-48];
        a[n-48]=a[n-49];
        a[n-49]=a[n-50];
        a[n-50]=a[n-51];
        a[n-51]=a[n-52];
        a[n-52]=a[n-53];
        a[n-53]=a[n-54];
        a[n-54]=a[n-55];
        a[n-55]=a[n-56];
        a[n-56]=a[n-57];
        a[n-57]=a[n-58];
        a[n-58]=a[n-59];
        a[n-59]=a[n-60];
        a[n-60]=a[n-61];
        a[n-61]=a[n-62];
        a[n-62]=a[n-63];
        a[n-63]=a[n-64];
        a[n-64]=a[n-65];
        a[n-65]=a[n-66];
        a[n-66]=a[n-67];
        a[n-67]=a[n-68];
        a[n-68]=a[n-69];
        a[n-69]=a[n-70];
        a[n-70]=a[n-71];
        a[n-71]=a[n-72];
        a[n-72]=a[n-73];
        a[n-73]=a[n-74];
        a[n-74]=a[n-75];
        a[n-75]=a[n-76];
        a[n-76]=a[n-77];
        a[n-77]=a[n-78];
        a[n-78]=a[n-79];
        a[n-79]=a[n-80];
        a[n-80]=a[n-81];
        a[n-81]=a[n-82];
        a[n-82]=a[n-83];
        a[n-83]=a[n-84];
        a[n-84]=a[n-85];
        a[n-85]=a[n-86];
        a[n-86]=a[n-87];
        a[n-87]=a[n-88];
        a[n-88]=a[n-89];
        a[n-89]=a[n-90];
        a[n-90]=a[n-91];
        a[n-91]=a[n-92];
        a[n-92]=a[n-93];
        a[n-93]=a[n-94];
        a[n-94]=a[n-95];
        a[n-95]=a[n-96];
        a[n-96]=a[n-97];
        a[n-97]=a[n-98];
        a[n-98]=a[n-99];
    }
    printf("array after rotation");
    for (i=0;i<n;i++){
        printf("%d",a[i]);
    }
    return 0;
}
```

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```
Last login: Sat Sep 19 14:07:17 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc rotate.c -opl
aditi@aditis-MacBook-Neo ~ % ./pl
enter the size of array 5
enter the elements of array 1 2 3 4 5
enter the value of k2
array after rotation52345
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int a[10][10];
    int rows,columns,i,j;
    printf("enter the no of rows");
    scanf("%d",&rows);
    printf("enter no of columns");
    scanf("%d",&columns);
    printf("enter matrix elements");
    for (i=0;i<rows;i++){
        for (j=0;j<columns;j++){
            scanf("%d",&a[i][j]);
        }
    }
    printf("matrix is \n");
    for (i=0;i<rows;i++){
        for (j=0;j<columns;j++){
            printf("%d",a[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

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```
aditi@aditis-MacBook-Neo ~ % nano printmatrix.c
aditi@aditis-MacBook-Neo ~ % gcc printmatrix.c -opn
aditi@aditis-MacBook-Neo ~ % ./pn
enter the no of rows2
enter no of columns3
enter matrix elements 1 2 3 4 5 6
matrix is
123
456
aditi@aditis-MacBook-Neo ~ % █
```



```
#include <stdio.h>
int main()
{
    int a[10][10];
    int rows,columns,i,j,sum=0;
    printf("enter number of rows\n");
    scanf("%d",&rows);
    printf("enter number of columns\n");
    scanf("%d",&columns);
    printf("enter matrix elements\n");
    for (i=0;i<rows;i++){
        for(j=0;j<columns;j++){
            scanf("%d",&a[i][j]);
            sum=sum+a[i][j];
        }
    }
    printf("sum of all elements=%d",sum);
    return 0;
}
```

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```
Last login: Sat Sep 19 14:35:31 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc sumofelements2d.c -opk
aditi@aditis-MacBook-Neo ~ % ./pk
enter number of rows 2
enter number of columns 3
enter matrix elements 1
2
3
4
5
6
sum of all elements=21
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int a[10][10],sum[10];
    int rows,columns,i,j;
    printf("enter rows");
    scanf("%d",&rows);
    printf("enter columns");
    scanf("%d",&columns);
    printf("enter matrix elements");
    for (i=0;i<rows;i++){
        sum[i]=0;
        for (j=0;j<columns;j++){
            scanf("%d",&a[i][j]);
            sum[i]=sum[i]+a[i][j];
        }
    }
    printf("sum of each row");
    for (i=0;i<rows;i++){
        printf("row %d=%d\n",i+1,sum[i]);
    }
    return 0;
}
```

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```
#include <stdio.h>
int main()
{
    int a[10][10], transpose[10][10];
    int rows, columns, i, j;
    printf("enter rows");
    scanf("%d", &rows);
    printf("enter columns");
    scanf("%d", &columns);
    printf("enter matrix elements");
    for (i=0; i<rows; i++){
        for (j=0; j<columns; j++){
            scanf("%d", &a[i][j]);
        }
    }
    for (i=0; i<rows; i++){
        for (j=0; j<columns; j++){
            transpose[j][i]=a[i][j];
        }
    }
    printf("transpose of matrix");
    for (i=0; i<columns; i++){
        for (j=0; j<rows; j++){
            printf("%d", transpose[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

```

Last login: Sat Sep 19 14:37:44 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc sumofrow.c
aditi@aditis-MacBook-Neo ~ % gcc sumofrow.c -opl
aditi@aditis-MacBook-Neo ~ % ./pl
enter rows 2
enter columns 3
enter matrix elements 1
2
3
4
5
5
sum of each rowrow 1=6
row 2=15
aditi@aditis-MacBook-Neo ~ % nano transpose.c
aditi@aditis-MacBook-Neo ~ % gcc transpose.c -opk
aditi@aditis-MacBook-Neo ~ % ./pk
enter rows 2
enter columns 3
enter matrix elements 1
2
3
4
5
5
transpose of matrix141869606976
251869607057
aditi@aditis-MacBook-Neo ~ % nano transpose.c
aditi@aditis-MacBook-Neo ~ % gcc transpose.c -opk
aditi@aditis-MacBook-Neo ~ % ./pk
enter rows 2
enter columns3
enter matrix elements
1
2
3
4
5
5
transpose of matrix14
25
36
aditi@aditis-MacBook-Neo ~ % █

```



```
#include <stdio.h>
int main()
{
    int a[10][10],b[10][10],sum[10][10];
    int rows,columns,i,j;
    printf("enter rows");
    scanf("%d",&rows);
    printf("enter columns");
    scanf("%d",&columns);
    printf("enter first matrix elements");
    for (i=0;i<rows;i++){
        for (j=0;j<columns;j++){
            scanf("%d",&a[i][j]);
        }
    }
    printf("enter second matrix");
    for (i=0;i<rows;i++){
        for (j=0;j<columns;j++){
            scanf("%d",&b[i][j]);
        }
    }
    for (i=0;i<rows;i++){
        for (j=0;j<columns;j++){
            sum[i][j]=a[i][j]+b[i][j];
        }
    }
    printf("sum of two numbers");
    for (i=0;i<rows;i++){
        for (j=0;j<columns;j++){
            printf("%d",sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

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```
Last login: Sat Sep 19 14:45:36 on ttys000
aditi@aditis-MacBook-Neo ~ % nano addmatrix.c
aditi@aditis-MacBook-Neo ~ % gcc addmatrix.c -o pl
aditi@aditis-MacBook-Neo ~ % ./pl
enter rows2
enter columns2
enter first matrix elements 1
2
3
4
enter second matrix 5
6
7
8
sum of two numbers68
1012
aditi@aditis-MacBook-Neo ~ % █
```

